

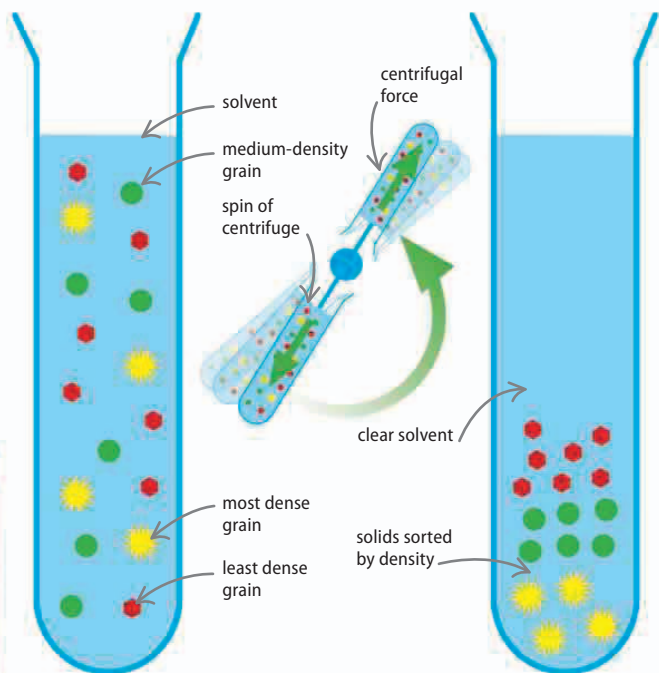
Centrifugal force

Another way to separate an uneven mixture is by using a centrifuge. In a suspension, the solid particles are often still too small to sink to the bottom under the pull of gravity alone. So the mixture is spun around at high speed, creating a centrifugal force that pushes the solid material down to the bottom of the test tube.

REAL WORLD

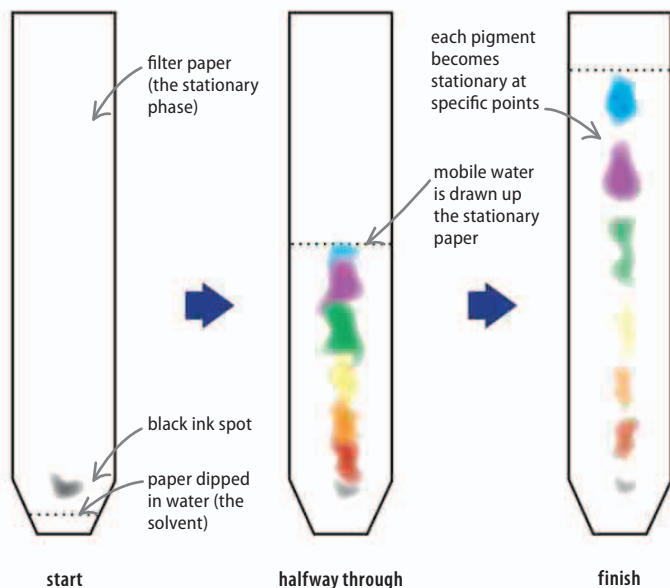
Butter churn

Butter is made by separating the solid fats from the liquid component in milk. This is done by mechanical disruption. The mixture is churned (spun) and this makes the blobs of fat stick together. The fat blobs get bigger and bigger until they separate from the water. The products from the churning are butter and buttermilk—a thinner, lower-fat liquid.



△ Sorting mixtures

The centrifugal force has the strongest effect on the densest particles in the mixture, so these move to the bottom fastest. This phenomenon can be used to sort suspended particles or grains—the densest ones form the lower layer, with successively less dense particles layered on top.



Chromatography

When the components of a mixture have the same particle sizes or similar boiling points, they are separated by chromatography. The mixture is dissolved in a solvent. The solvent, known as the mobile phase, is then drawn through a substance known as the stationary phase, often filter paper. The mobile phase moves forward, but each component in the mixture moves through the stationary phase at a different speed. As a result, each component becomes fixed in the stationary medium at a different point, forming separated samples of each substance.

◁ Separating black ink

Black ink is a mixture of colored pigments in water. These can be separated using chromatography. The word means “writing with color,” and the drop of ink forms bands of its individual color components.